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From knowledge to responsible actions: The roles of willingness to sacrifice and personal involvement in carry-out trash behavior

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Abstract

Active removal of self-generated litter in mountainous destinations has emerged as an important solution to tourism waste management. However, tourists' participation remains limited, and the underlying psychological mechanisms are not fully understood. Drawing on the low-cost hypothesis, this study extends the knowledge-attitude-behavior framework by incorporating willingness to sacrifice and personal involvement to explain proactive carry-out trash behavior. Based on 264 field questionnaires collected from tourists visiting a world-class mountain destination in eastern China, this study examines the psychological mechanisms underlying carry-out trash behavior. The results reveal a significant serial mediation process: environmental knowledge influences behavioral intention indirectly through tourists' attitude and their willingness to sacrifice personal convenience. This finding highlights the importance of subjective cost-benefit evaluation in translating positive attitude into action. Furthermore, personal involvement in everyday waste management negatively moderates the relationship between tourists' environmental knowledge and attitude toward carry-out trash behavior. Specifically, the positive effect of environmental knowledge on attitude is stronger among tourists with lower levels of personal involvement. By integrating willingness to sacrifice and personal involvement into the KAB model, this study advances understanding of tourists' environmentally responsible behavior in mountainous contexts and provides practical implications for designing targeted behavioral interventions to promote carry-out trash practices.

Keywords

Knowledge-attitude-behavior framework; Waste management; Willingness to sacrifice; Personal involvement; Leave no trace; Mountainous destination

1. Introduction

The rapid growth of mountain tourists has significantly increased the burden of waste management, posing a threat to the sustainable development of mountainous destinations (Romeo et al., 2021). Mountainous regions worldwide are dealing with a heavy burden of waste and environmental crisis due to excessive tourism flows (Kaseva & Moirana, 2010). “Leave no Trace”, a campaign which encourages visitors to proactively reduce waste and harm to the nature, have been widely advocated since the 1960s (Hu et al., 2019; Ng, 2023). Yet the initiative remains largely confined to a small cohort of dedicated outdoor practitioners, with its reach among the general tourist public still pending meaningful expansion.

Tourist environmentally responsible behavior (ERB) is seen as the critical means to address global climate and environmental change as well as promote sustainable development (Barr, 2003). Within the literature on waste-related ERBs, prior studies have investigated tourists’ immediate binning and resorting behavior at the point of consumption (Esfandiar et al., 2020; Zhong et al., 2024), while less attention has been paid to subsequent waste management behaviors such as carrying waste away for later disposal or reusing items (Raghu & Rodrigues, 2020). These later-stage behaviors require sustained effort, temporary storage and continued environmental commitment. Many destinations face challenges in collecting and transporting on-site tourism waste due to limited labor force and challenging terrain (UNEP, 2016). As a result, tourists’ proactive environmentally responsible behaviors, especially taking selfgenerated litter out of the zones, are quite valuable. Building on Principle 3 of “Leave No Trace” (Dispose of Waste Properly), this study focuses on tourists’ voluntary behavior of carrying waste out of core scenic areas, which we term carry-out trash behavior.

Tourists’ carry-out trash behavior in mountainous destinations poses a distinct challenge, as it is typically voluntary, non-mandatory, and enacted under strong physical and situational constraints (Matiiuk & Liobikienė, 2021). Performing this behavior requires tourists to recognize environmental consequences, interpret site-specific information, know how to carry out trash properly, and translate such understanding into concrete actions, rendering it a cognitively demanding rather than purely normative or habitual response (Zhang, 2023). Behavioral frameworks commonly applied in tourism research, such as the theory of planned behavior and norm-based models (Lam et al., 2025; Ng & Cheng, 2022), focus heavily on intentions, social norms, and moral obligations, but tend to overlook the role of knowledge. In contrast, the knowledge–attitude–behavior (KAB) framework conceptualizes environmentally responsible behavior as a sequential process in which knowledge constitutes the cognitive foundation for attitude formation and subsequent actions (Maleknia & Zamani, 2025). This process-oriented perspective aligns closely with the context-specific nature and knowledge/skill demands of tourists’ carry-out trash behavior in mountainous settings. Accordingly, this study adopts the KAB framework to examine how environmental knowledge translates into attitudes and actual carry-out trash behavior.

While the KAB framework provides a useful baseline for explaining how knowledge and attitudes influence environmentally responsible behavior, its explanatory power may be limited

in tourism contexts where situational constraints are salient (Gkargkavouzi et al., 2019). Inspired by the low-cost hypothesis (LCH) (Diekmann & Preisendörfer, 1998) and the growing interest in attitude-behavior gap (Dolnicar, 2020), recent studies increasingly recognize situational factors as critical supplements to models that focus solely on psychological variables (Miao & Wei, 2013; Schmitt et al., 2018). Tourists often behave differently from their everyday lives, such that environmental responsible practice routinely performed at home are not readily transferred to travel contexts (Bilynets & Knezevic Cvelbar, 2022; Juvan & Dolnicar, 2014). Moreover, mountainous tourism settings are characterized by heightened constraints, barriers and costs, making tourism ERBs more difficult to perform or sustain (Cleveland et al., 2020; Gaspar, 2013; Moghimehfar & Halpenny, 2016). As a result, tourists may endorse environmental values or norms without translating them into actual behavior (Gillani et al., 2021; Nieto-García et al., 2024). These insights suggest that explaining tourists' carry-out trash behavior requires not only accounting for cognitive and attitudinal process but also incorporating tourists' willingness to incur behavioral costs and their involvement with everyday waste management.

Building on the above observations, this study proposes an extended KAB framework by integrating willingness to sacrifice and personal involvement as two critical variables to better understand tourists' carry-out trash behavior. Empirical data were collected via a survey with 264 visitors in *Mount. Huangshan*. This study reveals the inherent difficulty of proactive ERB in mountainous tourism contexts and offers a new perspective for understanding how ERB unfolds beyond attitude and intention under constrained conditions. These insights also provide insights for policymaking and intervention to promote visitors' participation in carry-out trash behavior.

2. Literature review and hypothesis development

2.1. Leave no trace (LNT) and carry-out trash behavior

Leave No Trace (LNT) represents a series of widely recognized tourist environmental responsible behaviors (ERBs), aiming to foster visitors' care for nature during outdoor recreation and leisure activities (Shenyoputro et al., 2024). As visitation to remote natural areas increased since 1960s, the LNT ethic emerged as a systematic framework to minimize human impact. In 1994, the Leave No Trace Center for Outdoor Ethics was formally established, and it subsequently codified the widely recognized seven principles of LNT: (1)Plan ahead and prepare; (2)Travel and camp on durable Surfaces; (3)Dispose of waste properly; (4)Minimize campfire impacts; (5) Leave what you find; (6)Be considerate of other visitors and (7) Respect wildlife (Morton Turner, 2002). Aligning with academia concerns about tourists ERBs (Bilynets & Knezevic Cvelbar, 2022; Schönherr, 2024), LNT offers a set of concrete guidelines for enacting site-specific ERBs (Lee et al., 2013; Ramkissoon et al., 2013) in natural recreation settings (Ng, 2023; Shenyoputro et al., 2024). Extant research has predominantly relied on the theory of planned behavior to investigate the predictive roles of behavioral, normative and

control beliefs in shaping LNT behavioral intentions (Ng, 2023; Vagias et al., 2014). Additional factors, such as group identity, environment interpretation and environment background, have been incorporated as moderators of the relationships between antecedents and intentions (Shenyoputro et al., 2024; Wang, Zhang, Cao, Duan, et al., 2019; Wang, Zhang, Cao, Hu, et al., 2019; Wang et al., 2018). As a result, both the analytical frameworks and empirical conclusions of this literature closely align with those of broader studies on environmentally responsible behavior in tourism contexts (Aziz & Niazi, 2023; Fakfare et al., 2024; Gabarda-Mallorquí et al., 2024; Horng et al., 2014).

Principle 3 of LNT, "Dispose of waste properly" is critical to the sustainability of outdoor tourism destinations. The most important and iconic practice, "Pack it in, Pack it out", encouraging visitors to carry out all waste they generate from natural areas. Different from other common waste-related ERBs including sorting, recycling and binning (Esfandiar et al., 2020), the carry-out trash behavior requires visitors to plan ahead by identifying potential waste streams and learning how to properly manage each type of waste according to local ecological conditions and disposal constraints (Shenyoputro & Jones, 2025). Despite these distinctive features, research examining this specific LNT practice remains scarce (Lam et al., 2025; Shenyoputro & Jones, 2025). Existing studies mainly rely on intention-based self-reports and generally echo findings from the broader literature on environmentally responsible behavior. Consequently, prior research has not adequately addressed the practical challenges of carry-out trash behavior, which is highly autonomous and requires continuous effort throughout the trip (Lam et al., 2025). Moreover, insufficient knowledge about potential waste streams and appropriate disposal methods may constitute a critical barrier to the enactment of this behavior. However, previous studies have concentrated primarily on attitude while paying less attention to the role of cognitive and knowledge-based factors.

This study focuses a special LNT practice, carry-out trash behavior in mountain tourism context, which refers to the act of carrying self-generated waste throughout the visit and disposing of it appropriately after exiting the mountainous area. Existing LNT research has primarily concentrated on two types of scenarios. The first is commonly referred to as the *backcountry*, encompassing remote areas accessible only by hiking and lack formal environmental infrastructure, where visitor traces are often interpreted as direct disturbances to wilderness conditions (Dolman & Marion, 2022; Sharp et al., 2020). The second is the *front-country*, referring to developed trailheads and park gateways that are easily accessible from urban centers, characterized by high visitor density and the presence of visitor facilities (Xiao et al., 2018), where LNT behaviors play a critical role in mitigating tourism-induced environmental pressures. Different from these two types of scenarios, mountainous tourism destinations are characterized by high visitor volumes and constrained internal mobility, thus intensifying both the importance and practical difficulty of the carry-out trash behavior.

On the one hand, alpine regions are ecologically fragile systems in which tourism waste may exert prolonged impacts on soil, water, and wildlife habitats (Romeo et al., 2021). Under conditions of constrained vertical transportation, the longer waste remains *in situ*, the greater its

potential ecological risk. Although mature alpine destinations often provide on-site waste collection facilities, structural limitations in vertical transportation remain difficult to overcome, underscoring the importance of visitors actively carrying trash out of alpine tourism areas. On the other hand, mass tourism further intensifies waste management challenges in mountainous destinations (FAO/UNWTO, 2023; Romeo et al., 2021). Large volumes of visitors generate concentrated amounts of waste within short time periods, while waste transportation capacity frequently falls short of demand. Compared with experienced outdoor visitors, mass tourists often lack sufficient environmental awareness and practical waste management skills, while long-distance, cross-area carry-out trash behavior remains highly autonomous and subject to voluntary withdrawal.

Against this backdrop, this study seeks an appropriate theoretical foundation to develop a research framework and hypotheses that can more adequately explain carry-out trash behavior in such mountainous settings.

2.2. Theoretical foundation: Knowledge-Attitude-Behavior model

Given the characteristics of carry-out trash behavior in mountainous tourism destinations, we expect tourists' environmental knowledge regarding this specific behavior to play a pivotal role in shaping whether and how the behavior is enacted. Accordingly, this study adopts the knowledge-attitude-behavior framework as the theoretical foundation, which conceptualizes individual behavior as a cognitively grounded process (Baranowski et al., 2003; Tillyard & DeGennaro, 2019). Within this framework, environmental knowledge is not treated as a peripheral background factor but as an active driver shaping individuals' understanding of environmental consequences and appropriate responses (Kallgren & Wood, 1986). This knowledge-centered perspective has been widely adopted in fields such as environmental education and consumer research to inform behavior-oriented interventions (Dhir et al., 2021; Marcinkowski & Reid, 2019; Taufique et al., 2017).

Grounded in the core assumptions of the KAB framework, this study posits that tourists' accumulation of environmental knowledge fosters a more favorable attitude toward carry-out trash behavior, which subsequently motivates behavioral engagement. Moreover, in light of the well-documented attitude-behavior gap in ERB research (Huffman et al., 2014; Juvan & Dolnicar, 2014), we incorporate behavioral intention (Ajzen, 1985) as a key mediating construct between attitude and actual behavior. Carry-out trash behavior is therefore conceptualized as a multi-stage process encompassing knowledge acquisition, attitudinal evaluation, formation of behavioral intention and behavioral enactment. Accordingly, we put forth the following hypotheses:

H1a: Tourists' environmental knowledge positively influences their attitude toward carryout trash behavior in mountainous destinations.

H1b: Tourists' attitude toward carry-out trash behavior in mountainous destinations positively influences their behavioral intention to perform such behavior.

H1c: Tourists' behavioral intention to perform such behavior positively influences their actual carry-out trash behavior in mountainous destinations.

2.3. Extension to the KAB model: willingness to sacrifice

In mountainous tourism destinations, carry-trash out behavior represents a distinctive form of environmentally responsible behavior characterized by salient and unavoidable personal costs. Unlike routine waste disposal in everyday settings, carrying self-generated trash out of the touristic area requires tourists to physically carry waste over long distances, tolerate inconvenience, invest additional time and effort, and sometimes endure discomfort such as dirty items and extra weight. These tangible and immediate burdens render carry-out trash behavior a high-cost and context-dependent practice, whose enactment requires more than favorable attitudes alone.

To account for how tourists overcome these cost-related barriers, this study introduces willingness to sacrifice as a key psychological mechanism (Ribeiro et al., 2025). Willingness to sacrifice refers to "the extent to which individuals' decisions will take into account the wellbeing of the environment, even at the expense of immediate self-interest, effort, or costs" (Davis et al., 2011). Instead of a stable personal disposition, this study conceptualizes willingness to sacrifice as a context-specific and behavior-oriented psychological state, reflecting tourists' deliberate readiness to accept the concrete inconveniences and efforts associated with carry-out trash behavior (Han et al., 2017; Rahman & Reynolds, 2016).

Attitude-behavior research consistently demonstrates that favorable evaluations of a behavior increase individuals' readiness to overcome barriers to its performance (Passafaro, 2020). When tourists hold a favorable attitude toward carrying out trash, such as viewing it as important and beneficial, they are more likely to accept the associated costs and inconveniences. In other words, we predict that a positive attitude toward carry-out trash behavior will strengthen tourists' willingness to tolerate the physical burden, time investment, and logistical challenges inherent in this practice. Subsequently, the increased willingness to sacrifice will further enhance their behavioral intentions.

H2a: Tourists' attitude toward carry-out trash behavior positively influences their willingness to sacrifice personal convenience for carry-out trash behavior in mountainous destinations.

H2b: Tourists' willingness to sacrifice positively influences their behavioral intention to perform carry-out trash behavior in mountainous destinations.

From a process perspective, willingness to sacrifice captures a critical psychological transition between attitude and behavior intention (Lou & Li, 2023). Even when tourists hold positive environmental attitudes, engaging in carry-out trash behavior requires an additional evaluative step in which individuals assess whether they are willing to endure the specific physical and situational costs involved. More specifically, individuals first develop the attitudes, then consider the costs associated with the behavior and the constraints they may face within a given situational context and form a relatively stable behavioral intention. Accordingly, we further hypothesized that tourists' willingness to sacrifice functions as a mediator through which

favorable attitude toward carry-out trash behavior are translated into behavioral intention in the context of high-cost environmental actions.

H3: Tourists' willingness to sacrifice mediates the relationship between their attitude toward carry-out trash behavior in mountainous destinations and their behavioral intention to perform such behavior.

Extending this mechanism within the KAB framework, environmental knowledge provides the cognitive basis for forming favorable attitude, which subsequently shape tourists' willingness to accept situational costs. Therefore, attitude and willingness to sacrifice are expected to operate sequentially as mediators linking environmental knowledge to behavioral intention.

H4: Tourists' attitude and willingness to sacrifice sequentially mediate the relationship between environmental knowledge and behavioral intention to perform carry-out trash behavior in mountainous destinations.

2.4. Extension to the KAB model: Personal involvement

Environmental knowledge is widely recognized as an important antecedent of individuals' attitudes and ERBs (Chen et al., 2023; Han & Hyun, 2017; Lee & Moscardo, 2005), but its effects are conditional and vary across individuals (Hines et al., 1987). This is particularly salient in tourism contexts, where ERBs often differ substantially from everyday practices. In this study, carry-out trash behavior exemplifies a context-dependent, non-routine ERB unfamiliar to many mass tourists, which leads to an important question: under what conditions does environmental knowledge translate into positive attitudes in specific tourism contexts?

This study introduces personal involvement in everyday waste management as a key individual-level boundary condition that explains how environmental knowledge is activated and used in attitude formation. Personal involvement refers to the degree of perceived relevance, importance, and sustained psychological engagement that individuals attach to certain object or issues (Russell-Bennett et al., 2007; Zaichkowsky, 1985). Rather than reflecting behavioral frequency, personal involvement represents a cognitive-affective orientation that influences individuals' motivation to attend to, elaborate on, and integrate relevant information (Petty & Cacioppo, 2012). Prior studies indicate that higher levels of personal involvement are associated with stronger, more accessible and stable attitude (Goeckeritz et al., 2010; Luttrell & Sawicki, 2020). Tourists' involvement with environmental issues may condition how knowledge encountered *in situ* affects their evaluations of non-routine ERBs (Nascimento & Loureiro, 2024; Sarabia-Sánchez et al., 2014).

Attitude strength theory suggests that attitudes formed under high personal involvement tend to be stronger, more stable, and more resistant to change (Maio & Haddock, 2010; Petty & Cacioppo, 2012; Petty et al., 1997). Such attitudes are more likely to be grounded in identity commitment and internalized values rather than incremental cognitive inputs. Consequently, when involvement is high, attitudes become less susceptible to additional information. In the context of mountainous tourism, with higher involvement in everyday waste management tend

to have stronger and more stable attitude toward carry-out trash behavior, making them less dependent on situationally presented environmental knowledge. Conversely, tourists with lower involvement with waste-related issues have weaker pre-existing attitudes and rely more on environmental knowledge activated in the tourism context, such as signage, guides, or observation of others (Goeckeritz et al., 2010; Luttrell & Sawicki, 2020). Personal involvement thus conditions how environmental knowledge is elaborated and integrated into attitudes, justifying its role as a moderator specifically for the knowledge-attitude link.

H5: Personal involvement negatively moderates the relationship between environmental knowledge and attitude toward carry-out trash behavior in mountainous destinations, such that the effect of environmental knowledge on attitude is stronger for tourists with lower personal involvement and weaker for tourists with higher personal involvement.

2.5. Research model

Fig. 1 displays the proposed research model extending the KAB framework to carry-out trash behavior in mountainous tourism contexts. Environmental knowledge is treated as an antecedent of tourists' attitude, which in turn influence behavioral intention and actual behavior. Willingness to sacrifice is introduced as a mediator, capturing the evaluative process by which tourists weigh personal costs to translate attitude into intention. Personal involvement is a moderator of the knowledge-attitude link, reflecting how personal involvement in everyday waste management conditions the elaboration and activation of environmental knowledge. The detailed hypotheses are organized into five groups (H1a–H5), collectively specifying these direct, indirect, serial mediating, and moderating effects.

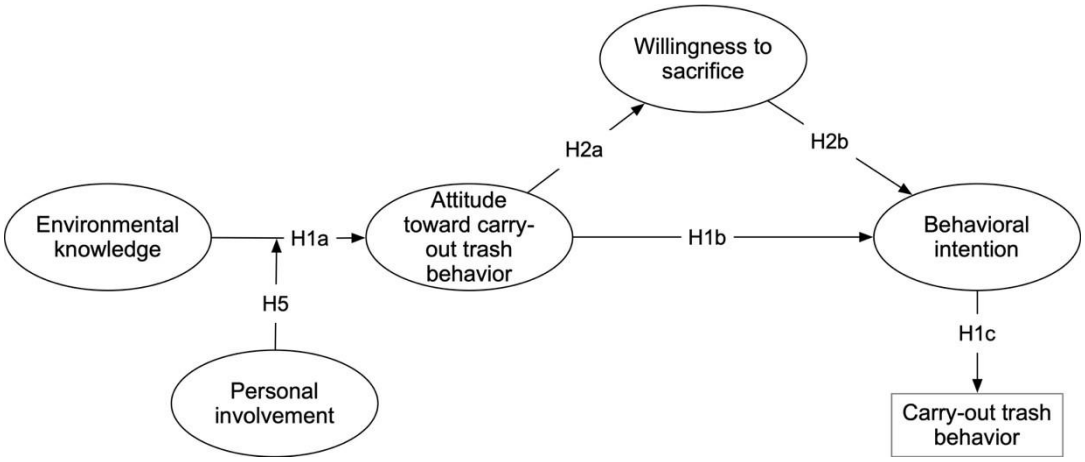


Fig. 1 Research model

Notes: Mediation paths for H3 and H4 are not illustrated in the figure; they are implied by the existing paths and tested statistically in the analysis.

3. Research Method

3.1. Research context

This study was conducted in Huangshan (https://hsgwh.huangshan.gov.cn/index.html), one of China's most iconic mountain destinations. Designated as a UNESCO World Heritage Site, Huangshan attracted over five million visitors in 2024 while also facing significant environmental challenges, particularly concerning tourist-generated waste. Official statistics indicated that tourism-related waste reached 4,400 metric tons in 2019, declined during the COVID-19 pandemic, and rebounded to more than 2,800 metric tons by 2024. The large influx of visitors and the associated waste continue to exert substantial pressure on local management capacity. To mitigate this challenge, an initiative was lunched in 2013 to encourage tourists to voluntarily carry their litter out of the high-altitude scenic zones. Within the scenic area, prominent signage reminds visitors to carry their waste down the mountain. A waste collection station is installed at the trail exit, and an incentive scheme has been introduced whereby tourists who bring down their litter can earn "green points" redeemable for merchandise (Cheng & Lv, 2021). However, evidence from the research team's long-term field observations in Huangshan and interviews with destination managers suggests these initiatives have fallen short of expectations, underscoring the need for a deeper understanding of the behavioral drivers behind tourists' carry-out trash behavior.

3.2. Measures

All measurement items (see Appendix A) were adopted from previously established scales and were contextually adapted to fit the present study on tourists' carry-out trash behavior in mountainous destinations. Minor modifications were made to item wording to improve clarity and contextual relevance, while preserving the original conceptual meaning of each construct. Environmental knowledge was measured using three items adopted from Li et al. (2018) and Wang et al. (2020). These items assess tourists' knowledge related to tourism waste from three aspects: general awareness of the environmental harm caused by tourism waste, understanding of the benefits of specific pro-environmental actions, and knowledge of practical ways to implement such actions during travel. Attitude towards carry-out trash behavior and behavioral intention were measured using established items in waste-related ERB researches (Ajzen, 1991; Ng, 2023; Vagias et al., 2014; Wang et al., 2020). Willingness to sacrifice, adopted from Mayerl and Best (2019), captures tourists' readiness to tolerate inconvenience, additional effort, or reduced comfort to perform carry-out trash behavior. Personal involvement measures individuals' perceived relevance of the environmental issue via three items from Wang et al. (2021). All measurement items were translated into Chinese using a back-translation procedure. Specifically, the original English items were first translated into Chinese and then independently translated back into English. The back-translated versions were compared with the original items to ensure semantic consistency. All items were measured using a five-point Likert scale ranging from 1 ("strongly disagree") to 5 ("strongly agree").

To capture tourists' self-reported carry-out behavior, respondents reported the quantities of waste generated during their visit and the quantities carried down when leaving the mountain. The items were recorded across the most common waste categories identified in the 2021 Mountain Waste Survey (MRI, 2022), including plastic bottles, metal cans, soft plastics, paper wrappings, disposable tableware, daily necessities, and plastic garbage bags. Based on field observations, tourist-generated waste was further distinguished by its source, namely items brought onto the mountain by tourists themselves and items purchased on the mountain. Concrete examples were provided for each category to enhance comprehension and reporting accuracy. Although self-reported measures may involve response bias (Huffman et al., 2014), they offer a meaningful proxy for actual behavior and are commonly used in environmental behavior research (Lange & Dewitte, 2019; Takeuchi et al., 2024).

At the end, participants voluntarily reported demographic information (i.e., gender, age, yearly income, and highest education level) and travel-related characteristics (e.g., travel companions, frequency of travel in general, frequency of visiting mountainous tourism destinations, and prior visiting experience in Huangshan).

Before data collection, the questionnaire was reviewed and refined through a pilot test involving four experienced researchers and three students with prior experience visiting mountainous destinations.

3.3. Data Collection and respondents' profile

The survey was conducted in the Huangshan Scenic Area between July 9 and July 11, 2023. A convenience sampling approach was employed, whereby 13 trained student assistants distributed questionnaires at three key locations along the main tourist routes: two cableway stations at the base of the mountain and the bus transfer center. These sites were selected because they allow tourists to temporarily stop and rest during their descent.

Data collection took place during two time periods, 11:00-12:00 and 16:00-18:00, which were identified as appropriate intervals when most tourists return from the summit and pause at these locations before leaving the scenic area. After excluding questionnaires with substantial missing data or consistently identical responses (indicating a lack of response attentiveness), a total of 264 valid questionnaires were retained for analysis, yielding an effective response rate of 87.7%.

As shown in Appendix B, approximately 53% of the respondents were male. Most respondents were under 40 years of age and held an undergraduate degree, indicating a relatively young and well-educated sample. Regarding annual income, about 40% reported earning between RMB 50,000 and RMB 200,000, while a similar proportion reported an income below RMB 50,000. In terms of travel companionship, half of the respondents traveled with friends, colleagues, or classmates, nearly 40% traveled with family members, and fewer than 10% traveled alone. In addition, most respondents reported infrequent visits to mountainous destinations, and more than 85% were first-time visitors to *Huangshan*.

3.4. Common method variance

In this study, multiple procedural and statistical remedies were employed to address potential common method variance (Esfandiar et al., 2020; Ribeiro et al., 2025). Procedurally, respondents were informed that the survey was conducted solely for academic purposes rather than to evaluate the appropriateness of their behavior, which helps reduce social desirability bias. In addition, following the recommendation of Jordan and Troth (2020), predictor and outcome variables were presented on separate pages of the questionnaire to psychologically separate the constructs and reduce respondents' tendency to provide consistent answers due to the measurement method rather than the constructs themselves. Statistically, two complementary diagnostic analyses were conducted. First, Harman's single-factor test indicated that no single factor accounted for the majority of the covariance among the measurement items (Podsakoff et al., 2003), suggesting that common method variance is unlikely to be a dominant concern (Agyeiwaah & Zhao, 2023; Xie et al., 2020). Second, a confirmatory factor analysis demonstrated that the theoretically specified five-factor measurement model (environmental knowledge, attitude, willingness to sacrifice, behavioral intention, and personal involvement) provided a significantly better fit to the data than an alternative single-factor model in which all items loaded onto one common factor ($\Delta\chi^2 = 1106.70$, $\Delta df = 10$, $p < 0.001$). This result suggests that the observed variance is unlikely to be explained by a single underlying source. Taken together, these procedural and statistical remedies suggest that common method variance is unlikely to pose a serious threat to the validity of the study's findings.

3.5. Analytical approach

The data analysis in this study consisted of three stages. The first stage involved establishing the measurement model to assess the reliability and validity of all constructs in the model. The second stage focused on examining the structural model and evaluating the overall model fit using multiple fit indices. The third stage utilized CB-SEM with R lavaan package in RStudio software (Lam et al., 2025; Rasul et al., 2025) to test the proposed research model and the corresponding hypotheses, including mediation effect of willingness to sacrifice and moderation effect of personal involvement.

4. Results

4.1. The self-reported carry-out trash behavior

Tourists' waste generation and carry-out trash behaviors are illustrated in Fig. 2. First, results indicate that approximately 80% of the waste generated during the visit originated from items carried onto the mountain by tourists themselves, with bottled beverages (32.2%) and daily necessities (23.6%) being the largest contributors. In contrast, about 20% of the waste came from items purchased on the mountain, among which bottled water was the dominant source (43.6%). Second, the ratio of waste that each tourist carried out relative to the total waste

they generated during the visit (hereafter referred to as the *"carry-out ratio"*) was calculated to represent tourists' actual carry-out trash behavior. On average, 52.3% of tourist-generated waste was carried out from the mountain, and nearly 57% of respondents carried out more than half of their waste. In terms of waste categories, daily necessities were most likely to be carried out, whereas plastic litter was the least likely. Third, respondents reported their preferred strategies for managing waste during the visit. Approximately 31.8% of tourists disposed of waste in onsite bins, 12.5% preferred to carry all waste out, and 53.4% chose a combination of on-site disposal and carry-out trash.

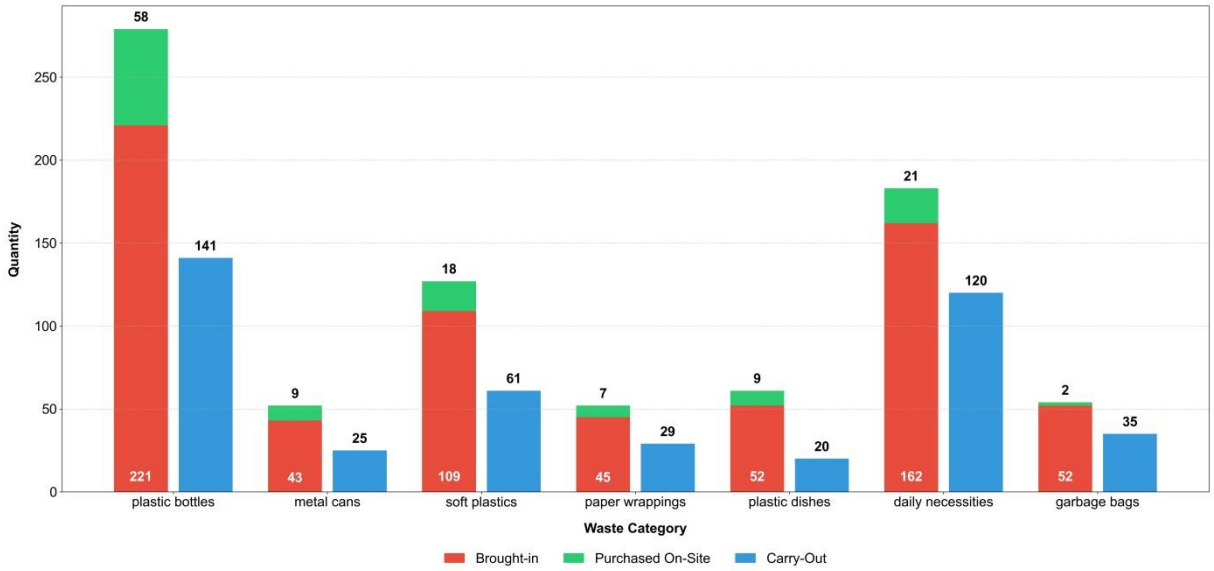


Fig. 2 Consumption sources and carry-out of tourism waste

4.2. Data reliability and validity

The confirmatory factor analysis (CFA) was performed to evaluate the validity of the model including five latent variables. With a maximum likelihood estimation method, the results of the CFA (χ^2=140.32, *df*=94, $p < 0.001$, χ^2/df = 1.493, RMSEA = 0.043, SRMR = 0.0441, GFI = 0.939, CFI = 0.985, NFI = 0.956, RFI = 0.944, TLI = 0.981, IFI = 0.985) are acceptable based on the criteria suggested by Hu and Bentler (1999). In other words, the proposed model appropriately fitted to the data. As reported in Appendix C, the factors loadings of all five constructs are above 0.6, and the Cronbach's values are above 0.7. The CR values ranged from 0.802 to 0.946, exceeding the recommended threshold of 0.6 (Bagozzi & Yi, 1988). Thus, the internal consistency was acceptable among multi-items for each variable.

Convergent validity was tested by standardized factor loadings of 16 observed variables varied from 0.616 to 0.94 (p <0.001), exceeding the recommended threshold of 0.6 (Chin et al., 2008). The average variance extracted (AVE) of five latent variables varied from 0.578 to 0.855, exceeding the minimum criteria of 0.50 (Fornell & Larcker, 1981), as shown in Tab. 1. The results supported the good convergent validity of latent variables. Moreover, the square root of AVE was greater than the correlation coefficients between any pair of latent variables (see Appendix C), indicating that the discriminant validity of each latent variable was satisfied.

Taken together, reliability, convergent validity, and discriminant validity of the measurement model were confirmed.

Tab. 1 The discriminant validity of latent variables

	Behavioral intention	Willingness to sacrifice	Attitude toward carry-out trash behavior	Personal involvement	Environmental knowledge
Behavioral intention	**0.891**				
Willingness to sacrifice	0.835	**0.925**			
Attitude toward carry-out trash behavior	0.491	0.502	**0.787**		
Personal involvement	0.588	0.541	0.367	**0.851**	
Environmental knowledge	0.38	0.38	0.577	0.41	**0.761**
AVE	0.794	0.8558	0.6192	0.7239	0.5789

Notes:（1）*Diagonal values indicated the square root of AVE of each latent variable; (2) Underneath of diagonal indicated the correlation matrix of latent variables.*

4.3. Hypothesis and model testing

Following the criteria suggested by Hu and Bentler (1999), the structural model in this study had a satisfactory fit (χ^2/df = 1.563, RMSEA = 0.046, SRMR = 0.052, CFI = 0.984,TLI = 0.980). As illustrated in Tab. 2, environmental knowledge significantly predicted attitude toward carry-out trash behavior ($p < 0.001$, $t = 7.143$). Attitude positively influenced both willingness to sacrifice ($p < 0.001$, $t = 7.944$) and behavioral intention ($p < 0.05$, $t = 1.994$). Willingness to sacrifice was the strongest predictor of behavioral intention ($p < 0.001$, $t = 13.818$), which in turn significantly predicted actual carry-out trash behavior ($p < 0.001$, $t = 4.355$). Taken together, H1a-H1c, H2a-H2b were all supported and the KAB model is accepted. The model demonstrates good explanatory power, accounting for 34% of the variance in attitude, 26% in willingness to sacrifice, and 58% in behavioral intention, indicating moderate to strong predictive validity in the context of tourist behavior environmental responsible behavior consistent with prior tourism studies (Chen & Tung, 2014; Han et al., 2010).

Tab. 2 Path analysis of the structural model

Hypothesis (Relation)	β	**t value**	**p value**	**Decision**
H1a (Environmental knowledge → Attitude toward carry-out trash behavior)	0.585	7.143	< 0.001	Accepted
H1b (Attitude toward carry-out trash behavior → Carry-trash-out intention)	0.101	1.994	0.046	Accepted
H1c (Carry-trash-out intention → Carry-out trash behavior)	0.270	4.355	< 0.001	Accepted
H2a (Attitude toward carry-out trash behavior → Willingness to sacrifice)	0.510	7.944	< 0.001	Accepted

H2b (Willingness to sacrifice → Carry-trash-out intention)	0.786	13.818	< 0.001	Accepted

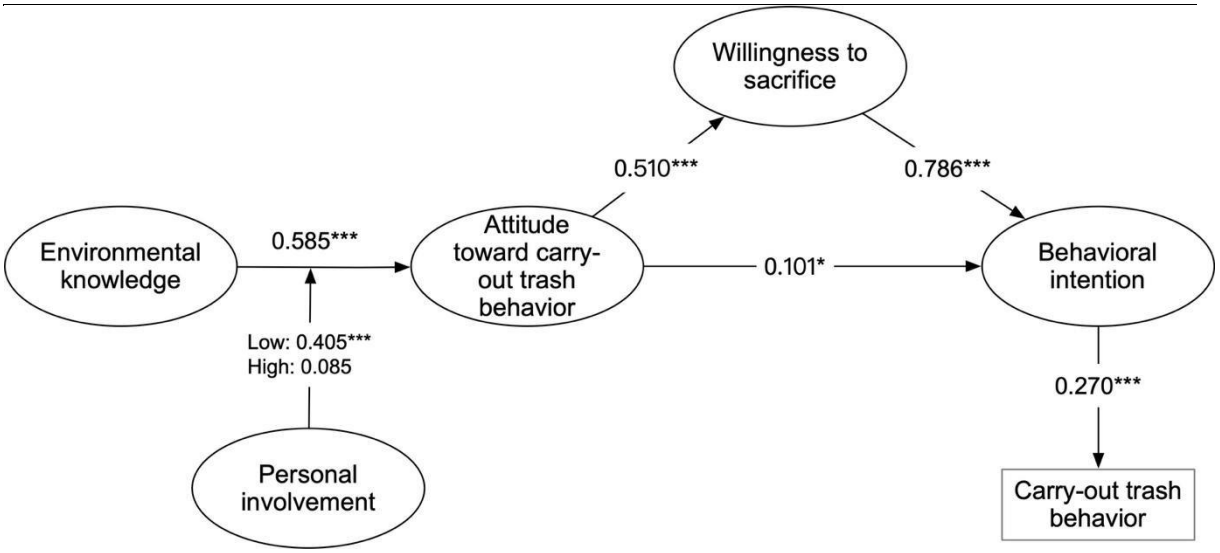


Fig. 3 Results of the model

Notes: (1) the moderation effect of personal involvement in low and moderate level are significant according to the bootstrapping 95% confidence intervals, with gender, age and time to mountainous areas are controlled;

*(2) ***p<0.001, **p<0.01, *p<0.05.*

4.4. The mediation effects analysis

The bias-corrected bootstrapping (5,000 resamples, 95% confidence intervals) is used to test the potential mediation effects. As shown in Tab. 3, although the hypothesized structural model specifies a sequential pathway from environmental knowledge to behavioral intention via attitude and willingness to sacrifice, direct paths (environmental knowledge → willingness to sacrifice, environmental knowledge → behavioral intention) were temporarily included during testing to assess whether the proposed simple mediation and serial mediation exist. Following recommended practices (Hayes, 2017), the non-significance of these direct effects, coupled with significant indirect effects provides evidence for full mediation. Specifically, attitude towards carry-out trash behavior mediates the relationship between environmental knowledge and willingness to sacrifice (B = 0.342, 95% CI = [0.1672, 0.35825]), and willingness to sacrifice mediates the relationship between attitude and behavioral intention (B = 0.5364, 95% CI = [0.4298, 0.8850]), supporting H3. Further, a serial mediation effect was confirmed that environmental knowledge positively influences behavioral intention via attitude and willingness to sacrifice (B = 0.3798, 95% CI = [0.2173, 0.6010]), supporting H4.

In fact, tourists visiting Huangshan are frequently exposed to information about the environmental harm of littering, practical methods for packing and carrying out trash, and strong promotional messages encouraging them to take their waste downhill. However, the verified pathway in this study clearly demonstrates the difficulty of carry-out trash behavior. Environmental knowledge does not directly translate into behavioral intention. Instead, it must

first foster a positive attitude toward the behavior, which then leads tourists to consciously evaluate and accept the personal sacrifices involved before forming a firm intention to act. It is precisely this multi-stage transformation, from knowledge to attitude, from attitude to willingness to sacrifice, and finally to behavioral intention, that helps explain why many tourists abandon the practice midway or fail to fully implement it, despite being aware of its importance.

Tab. 3 Mediation effect analysis

Model	Effect	Path	Unstd_Est	Std_Est	LLCI	ULCI	Significant
EK→WTS	**IE**	**via AT**	**0.3422**	**0.2441**	**0.1672**	**0.5825**	**Yes**
	DE	EK → WTS	0.1878	0.1340	-0.0693	0.5140	No
	TE	Total	0.5300	0.3781	0.2999	0.8754	Yes
AT→BI	**IE**	**via WTS**	**0.6195**	**0.3948**	**0.4298**	**0.8850**	**Yes**
	DE	AT → BI	0.1519	0.0968	-0.0615	0.4604	No
	TE	Total	0.7714	0.4916	0.4817	1.1914	Yes
EK→BI (serial)	**IE**	**via AT → WTS**	**0.3798**	**0.2440**	**0.2173**	**0.6010**	**Yes**
	DE	EK → BI	0.1267	0.0814	-0.0113	0.2949	No
	TE	Total	0.5065	0.3254	0.2919	0.7945	Yes

Notes: (1) IE= indirect effect, DE= direct effect, TE= total effect; (2) EK: environmental knowledge, AT: attitude toward carry-out trash behavior, WTS: willingness to sacrifice, BI: behavioral intention

4.5. The moderating analysis of personal involvement

A continuous moderation model was used to examine the moderating effect of personal involvement (PI) on the relationship between environmental knowledge (EK) and attitude toward carry-out trash behavior (AT) (Hayes, 2017). Composite scores were computed by averaging item responses. EK and PI were mean-centered prior to creating the interaction term to reduce multicollinearity. Gender, age, and time management were included as control variables. As expected, the interaction between EK and PI was statistically significant (b = 0.192, SE = 0.092, t = -2.09, p = .037, 95% CI = [–0.373, –0.012], HC3 robust standard errors), with an effect size (ΔR^2 = 0.033) acceptable in behavior research (Aguinis et al., 2005). Simple slope analyses (show in Fig. 4) revealed that EK positively predicted AT at low levels of PI (M1SD; b = 0.405, p < .001). However, at high levels of PI (M+1SD), the effect of EK was no longer significant (b = 0.085, p = .294). Johnson–Neyman analysis further indicated that EK significantly predicted AT only when PI was below 0.574. Taken together, H5 was supported.

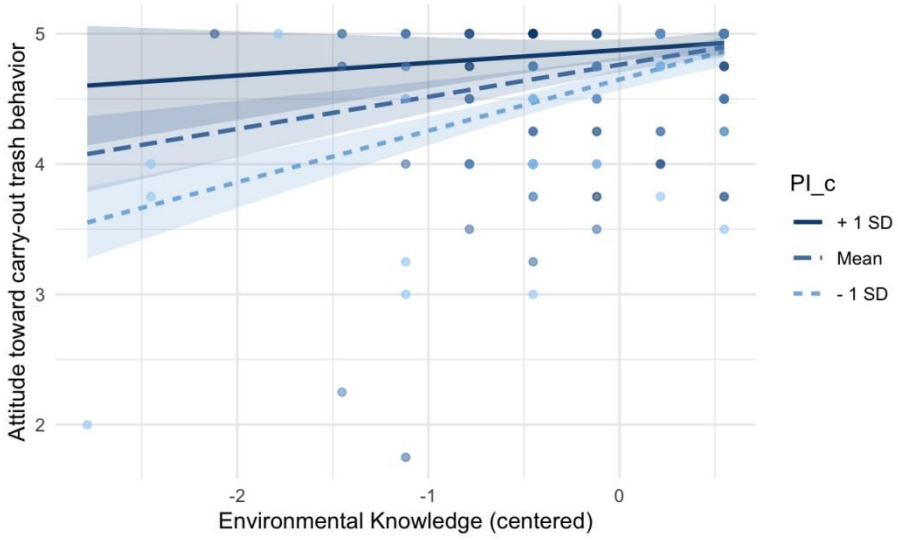


Fig. 4 Moderating effect of personal involvement

Notes：*EK=environmental knowledge, PI=personal involvement*

5. Discussion and conclusions

5.1. Conclusions and theoretical implications

Carry-out trash in mountainous tourism destinations represents an important environmental initiative, yet it remains difficult to secure broad tourist participation. Grounded in the authentic context of mountainous tourism, this study extends the knowledge–attitude–behavior framework to uncover the underlying mechanism driving carry-out trash behavior. By incorporating willingness to sacrifice and personal involvement into the model, the study deepens understanding of how tourists translate environmental knowledge into responsible actions. The findings illuminate the practical constraints and psychological trade-offs tourists experience when engaging in carry-out trash behavior in mountainous tourism settings and identify key leverage points for promoting such behavior. Overall, this research makes several important contributions to the literature on tourist environmentally responsible behavior (ERB).

First, this study extends the application of the knowledge-attitude-behavior (KAB) framework within the context of tourist environmentally responsible behavior (ERB). By situating the analysis in a mountainous tourism setting, the study re-conceptualizes the roles of environmental knowledge, attitude, and personal involvement, which have frequently appeared in prior ERB and waste-related behavior research. Unlike the large body of research grounded in the theory of planned behavior (TPB) emphasizing the role of social norms, this study suggests that carry-out trash behavior is less driven by externally imposed normative pressure and more reflective of a cognitive-behavioral alignment between individuals and the tourism environment. In mountainous destinations, there is no explicit or enforceable social norm mandating tourists to carry their waste out of core scenic areas, making internal cognitive factors particularly salient. Inspired by Leave No Trace Principle 3, this study highlights situational

environmental knowledge as a foundational driver of proactive environmental actions (Hu et al., 2018). Importantly, the identified serial mediation effect reveals that environmental knowledge influences behavioral intention through a sequential process by fostering a positive attitude and strengthening willingness to sacrifice, thereby clarifying the psychological mechanism through which knowledge is translated into responsible action.

Second, this study highlights the importance of incorporating behavioral costs and tradeoff process into the examination of tourists' ERB in mountainous destination by integrating willingness to sacrifice into the KAB model. Unlike low-effort pro-environmental actions, carry-out trash behavior often involves tangible physical inconvenience and psychological cost. Drawing on the low-cost hypothesis (Diekmann & Preisendörfer, 2003) and the notion that individuals engage in subjective cost-benefit trade-offs when making environmental decisions (Passafaro, 2020), this study conceptualizes willingness to sacrifice as a key internal mechanism reflecting tourists' readiness to accept personal costs for environmental responsibility (Chen et al., 2022). Empirically, willingness to sacrifice emerges as a significant mediator linking attitude to behavioral intention, thereby clarifying how positive evaluations are translated into actions under conditions of perceived cost. While prior studies have acknowledged situational constraints and normative pressures in shaping tourist ERBs (Brown et al., 2010; Wang, Zhang, Cao, Hu, et al., 2019), this study advances the literature by explicitly modeling the internal psychological trade-off process that underlies responsible behavior in tourism contexts. It highlights that tourists' engagement in ERB, particularly in mountainous destinations, depends on favorable attitudes translated into behavioral intention through a willingness to accept personal sacrifice.

Third, this study identifies personal involvement as a critical boundary condition that shapes the effectiveness of environmental knowledge in driving tourist ERBs. Rather than exerting a uniform influence, environmental knowledge operates differently depending on the level of personal involvement in everyday waste management, challenging the mainstream view that environmental knowledge always positively influences attitude (Paço & Lavrador, 2017). In our research, environmental knowledge exerts a stronger influence among tourists with lower levels of everyday involvement, whereas its effect diminishes among those who are already highly engaged in waste-related practices in daily life. This moderating pattern suggests that highly involved individuals possess relatively internalized and stable attitudes that are less dependent on situational knowledge in tourism contexts. In contrast, tourists with lower everyday involvement rely more heavily on external environmental knowledge inputs when evaluating carry-out trash behavior during travel. In this sense, personal involvement does not function as a cognitive amplifier, but rather as a stabilizing force that reduces the marginal contribution of additional knowledge.

5.2. Practical implications

This study offers practical insights for promoting carry-out trash behavior in nature-based tourism destinations. For ecologically fragile yet heavily visited areas, such as mountainous

regions, glaciers and protected natural reserves, encouraging such waste management practice is critical to alleviating environmental pressures and enhancing the efficiency of tourism waste disposal.

First, destination authorities should move beyond abstract environmental narratives and place greater emphasis on the dissemination of actionable and context-specific environmental knowledge. While general sustainability disclosures and political agendas (e.g., ecological civilization) are frequently communicated through infrastructure design and social media campaigns, they often lack guidance on what tourists can concretely do in specific tourism settings. Our findings suggest that tourists' engagement in carry-out trash behavior can be enhanced by providing clear and practical instructions, such as which items are encouraged or prohibited, what types of waste should be carried out, how waste can be packaged efficiently, and how physical burdens can be minimized. Delivering such knowledge at key on-site locations, particularly resting areas and camping sites, can directly shape tourists' behavioral decisions by reducing uncertainty and perceived effort.

Second, destination authorities should explicitly facilitate the transfer of environmentally responsible behaviors from daily life to tourism contexts. Many tourists are already accustomed to waste sorting, collection and centralized disposal in their everyday environments. However, tourism experiences often disrupt these routines. Instead of predominantly promoting the attractiveness and popularity of destinations, managers are encouraged to communicate the environmental challenges faced by local ecosystems, thereby positioning tourists as capable and responsible actors rather than passive consumers. By framing carry-out trash behavior as a continuation of familiar daily practices rather than exceptional requirement, destinations may foster tourists' positive attitudes toward such behavior and strengthen their intention to engage in it during travel.

Finally, greater attention should be paid to enhancing tourists' willingness to sacrifice through realistic and empathetic communication strategies. Rather than framing carry-out trash behavior as an occasional voluntary initiative, destination managers could proactively publicize this behavior through widely used social media platforms, especially prior to peak travel periods. Importantly, communication should highlight the real difficulties of waste transportation in mountainous environments, the potential harm of accumulated waste to wildlife, the physical demands faced by sanitation workers. Our field observations indicate that some tourists begin to reassess the value of carrying out their waste after witnessing the labor-intensive process of manual waste transportation by workers. Moreover, professional and transparent environmental knowledge dissemination can help correct tourists' overestimation of behavioral costs, thereby increase acceptance of personal sacrifice and foster sustained carry-out trash behavior.

5.3. Limitations and future research

Several limitations of this study should be acknowledged, which also point directions for future research. First, this study relied on a recall- and self-report-based approach to capture

tourists' actual carry-out trash behavior, given the limited feasibility of directly observing and tracking individual behaviors at a large and complex destination. Future research may consider smaller destinations, where emerging techniques such as field observation, manual weight and synchronous narrative interviews to measure real behavior (Viglia et al., 2024). Second, although this study extended the knowledge–attitude–behavior (KAB) model by incorporating willingness to sacrifice and personal involvement, it focused on a knowledge-driven explanatory framework. Other factors, such as environmental self-identity, the new ecological paradigm (Landon et al., 2018), and perceived destination trust and reputation (Aziz & Niazi, 2023), may complement or interact with this framework in shaping tourists' ERBs and warrant further investigation. Third, the sample was dominated by young, highly educated, first-time visitors, which may limit the generalization of the findings to other visitor segments or mountainous destinations. However, this sample structure broadly reflects the actual visitor composition of the study site, where younger and first-time tourists constitute a substantial proportion of total visitation and repeat visitation remains relatively low. Future research could examine more diverse visitor profiles and destinations to assess the robustness of the findings across different contexts. Finally, the cost-based extended KAB framework needs validation in other cultural contexts. Willingness to sacrifice is a particularly salient psychological driver in the Chinese context, where tourists often view waste management as the destination's responsibility. In such settings, trash carry-out behavior is more likely to depend on individuals' readiness to incur personal costs for collective outcomes. By contrast, in Western contexts where norms of individual environmental responsibility are more institutionalized, proenvironmental actions may be more strongly guided by internalized moral obligations.

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